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ORIGINAL PAPER

Genomic and transcriptomic analyses match medulloblastoma mouse models to their human counterparts

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Abstract Medulloblastoma is a malignant embryonal brain tumor with highly variable outcome. In order to study the biology of this tumor and to perform preclinical treatment studies, a lot of effort has been put into the generation of appropriate mouse models. The usage of these models, however, has become debatable with the advances in human medulloblastoma subgrouping. This study brings together multiple relevant mouse models and matches genetic alterations and gene expression data of 140 murine tumors with 423 human medulloblastomas in a global way. Using AGDEX analysis and *k*-means clustering, we show that the *Blbp-cre::Ctnnb1*$(ex3)^{Fl/+}$*Trp53*$^{Fl/Fl}$ mouse model fits well to human WNT medulloblastoma, and that, among various *Myc*- or *Mycn*-based mouse medulloblastomas, tumors in *Glt1-tTA::TRE-MYCN/Luc* mice proved to be most specific for human group 3 medulloblastoma. None of the analyzed models displayed a significant match to group 4 tumors. Intriguingly, mice with *Ptch1* or *Smo* mutations selectively modeled SHH medulloblastomas of adulthood, although such mutations occur in all human age groups. We therefore suggest that the infantile or adult gene expression pattern of SHH MBs are not solely determined by specific mutations. This is supported by the observation that human medulloblastomas with *PTCH1* mutations displayed more similarities to *PTCH1* wild-type tumors of the same age group than to *PTCH1*-mutated tumors of the other age group. Together, we provide novel insights into previously unrecognized specificity of distinct models and suggest these findings as a solid basis to choose the appropriate model for preclinical studies on medulloblastoma.

Keywords WNT · SHH · Medulloblastoma · Gene expression profile · Mutations · Mouse model

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Introduction

For decades, medulloblastoma was recognized as a malignant embryonal brain tumor with particularly poor prognosis. Affected patients were treated with surgery, chemotherapy and—depending on their age—radiotherapy. In the era of personalized medicine, this image has changed dramatically. Medulloblastoma is now acknowledged as a group of at least four major tumor types that significantly differ from each other with respect to genetics, epigenetics, gene expression, histology, and clinical outcome [17, 32, 33]. This knowledge requires stratified therapeutic approaches that refer to each tumor subgroup and to each tumor's genetics, which can vary even within one subgroup. Therefore, the use of mouse models to study a tumor's biology or to perform preclinical trials has to be carefully adapted to the tumor subgroup that is planned to be investigated or targeted [22]. *Ptch1*$^{+/-}$ mice, for instance, that develop medulloblastoma due to the constitutive activation of the Sonic hedgehog (Shh) pathway [9], have to be considered unsuitable for investigations that do not refer to medulloblastoma of the SHH subgroup. Hence, approximately 70 % of human medulloblastoma cases are not covered by this model [17]. More strictly, *Ptch1*$^{+/-}$ mice may even be unsuitable to model medulloblastoma that do not carry *PTCH1* mutations, which account for approximately 85 % of any medulloblastomas [14]. Meanwhile, a variety of additional transgenic mice has been generated that were suggested as models for 'medulloblastoma'. However, until recently, neither the technology nor a large enough cohort size of thoroughly analyzed human medulloblastoma samples was available to comprehensively categorize and validate the generated mouse models with respect to the growing diversity of human medulloblastoma.

This study brings together genome-wide sequencing data and gene expression profiles from 47 and 423 human medulloblastomas, respectively. These data are used as a reference to precisely match existing mouse models to subgroups of the human disease. This said, we acknowledge the commonly accepted four molecular groups of human medulloblastoma (WNT, SHH, Group 3 and Group 4) [47], but we also take into account newly identified subgroups within the cohort of SHH patients, i.e., SHH tumors that arise in infants and SHH tumors that arise in adults [31]. Based on our approach, this study indicates a substantial lack of reliable models for Group 4 medulloblastoma, but reveals surprising details on a number of genetically engineered mice that model human SHH medulloblastoma. Results from whole genome sequencing of mouse and human tumors demonstrate that the genetic make-up is not the single driver of a tumor's gene expression profile, which determines molecular medulloblastoma classification.

Materials and methods

Human tissue samples

Snap frozen human tissue samples were reviewed by experienced neuropathologists and examined for tumor content. Samples were excluded from analysis when <70 % of the sample contained tumor cells. Only primary medulloblastoma samples were included in the analysis. RNA and DNA were isolated using Trizol (Invitrogen, Carlsbad, CA, USA). RNA purification was performed using the RNeasy mini kit (Qiagen, Germantown, USA). DNA and RNA quantity and quality were determined by spectrophotometry (Nanodrop, Wilmington, USA) and microfluidics-based electrophoresis (Agilent 2100 Bioanalyzer, Agilent, Palo Alto, USA).

Gene expression analysis

423 human medulloblastoma samples and 18 human cerebellar tissue control samples were profiled on Affymetrix HGU133 plus 2.0 Genechips according to the manufacturer's instructions (Affymetrix Inc. Santa Barbara, USA). Data were normalized using the MAS 5.0 algorithm (Affymetrix Inc. Santa Barbara, USA). Medulloblastoma samples were classified into molecular subgroups using TM4 Microarray Software Suite (MeV v4.8; Dana-Farber Cancer Institute, Boston, MA, USA) by unsupervised hierarchical clustering using the Pearson correlation metric (average linkage) and bootstrapping analysis of high-standard deviation genes. Human gene expression data combined a cohort of data generated by ourselves and publicly available data with GEO Series accession numbers GSE49243, GSE12992, GSE10327 and GSE37418 [6, 18, 19, 39].

Primary murine medulloblastoma samples of *Blbp-cre::Ctnnb1(ex3)*$^{Fl/+}$*Tp53*$^{Fl/+}$ [8], *Nestin-creER*^{T2}*::Ptch1*$^{Fl/Fl}$ [29], *Math1-creER*^{T2}*::Ptch1*$^{Fl/Fl}$ [29], *Ptch1*$^{+/-}$ [9], *Nestin-cre::Cdkn2c*$^{-/-}$ *Trp53*$^{Fl/Fl}$ [15], *Cdkn2c*$^{-/-}$ *Ptch1*$^{+/-}$ [48], *MycDNTrp53* (*Prom1* + *Lin* + stem cells) [37], *MycDNTrp53* (*Prom1* + *Lin* + progenitors) [37], *Cdkn2c*$^{-/-}$ *Trp53*$^{-/-}$ *mMyc* [15], *Cdkn2c*$^{-/-}$ *Trp53*$^{-/-}$ *hMyc* [15, 54], and *Cdkn2c*$^{-/-}$ *Trp53*$^{-/-}$ *MycN* mice [15] as well as murine cerebella and granule neuron precursors were profiled using the Affymetrix Mouse Genome 430 2.0 Array, and cel files were assessed through GEO Series accession numbers GSE50824, GSE24628, GSE34126, GSE33199, GSE11859, GSE19534, and GSE10246 [21]. We previously profiled primary cerebellar and cochlear medulloblastomas from *Math1-cre::SmoM2*$^{Fl/+}$, *hGFAP-cre::SmoM2*$^{Fl/+}$, *Tlx3-cre::SmoM2*$^{Fl/+}$, and *Olig2-tva-cre::SmoM2*$^{Fl/+}$ mice on the Affymetrix Mouse Genome 430 2.0 platform [10, 41]. Gene expression data of *Ptch1*$^{+/-}$*Trp53*$^{-/-}$ and *Sufu*$^{+}$

$^{/-}Trp53^{-/-}$ medulloblastomas (Affymetrix MOE 430A/B platform) were kindly provided by Drs. Lee and McKinnon [24, 51]. Cerebella and granule neuron precursors as controls on the Affymetrix MOE 430A/B platform were assessed through GEO Series accession numbers GSE9299 and GSE6463 [30]. Gene expression data of preneoplastic $Ptch1^{+/-}$ lesions and primary $Ptch1^{+/-}$ medulloblastoma samples were generated with the Affymetrix Murine Genome U74A Version 2 Array and assessed through the GEO Series accession number GSE2426 [36]. All data were normalized with the MAS 5.0 algorithm using the Affymetrix expression console. Gene expression data of medulloblastomas from *Glt1-tTA::TRE-MYCN/Luc* mice and corresponding controls were generated on the Affymetrix Mouse Exon 1.0 ST Array and were assessed through GEO Series accession number GSE36594 [46]. Data were normalized on gene level by the RMA Sketch algorithm using the Affymetrix expression console.

Integrating gene expression platforms

In order to ensure that in a platform each gene was only represented once, the maximum variance approach was used to select a probe-set. For each probe-set of a gene, the average expression and variance over all samples in a platform were calculated and the probe-set with the highest variance was chosen to represent this gene in the analyzed data. In order to unify gene expression measures from multiple platforms, data from each batch on each platform were normalized and summarized separately resulting in gene expression estimates for each sample and gene on each platform. Relative gene expression values were calculated the following way: from each gene expression value the mean gene expression value across all samples was subtracted and the result was divided by the standard deviation across all samples. In order to remove known batch effects, the ComBat function within the surrogate variable analysis (sva) package was used, whenever applicable [13, 25]. In order to compare and select reported genes between different platforms, whenever available, the Affymetrix Good Match spreadsheets were used (www.affymetrix.com), else the Chip comparer tool using NCBI-HomoloGene (http://chipcomparer.genome.duke.edu) was applied. Only orthologue genes that were reported on all the analyzed platforms were included in the analysis. This resulted in 10,333 genes for the comparison of the Affymetrix HGU133 plus 2.0, the Affymetrix 430 2.0, the Affymetrix MOE 430A/B and the Affymetrix Mouse Exon 1.0 ST Array and in 7329 gene level estimates for the comparison of the Affymetrix HGU133 plus 2.0 and the Affymetrix Murine Genome U74A Version 2 Array.

Unsupervised hierarchical clustering, k-means clustering and supervised analysis

Unsupervised hierarchical clustering was performed using the Pearson correlation metric (average linkage) using the TM4 Microarray Software Suite (MeV v4.8; Dana-Farber Cancer Institute, Boston, MA, USA). *K*-means clustering was performed with the TM4 Microarray Software Suite using the Pearson correlation metric and up to 150 permutations with *k*-values based on the established number of clusters in human medulloblastomas [31, 47]. For supervised analysis, significantly expressed genes in each subgroup were identified by comparing each cluster with all other tumors. The student *t* test for two groups to be compared, or else ANOVA, corrected for multiple testing using the adjusted Bonferroni correction and an alpha value of 0.05, was applied.

Agreement of differential expression (AGDEX)

In order to compare the transcriptome of human and murine tumor samples on different platforms the agreement of differential expression (AGDEX) statistic was applied as previously published [12, 38]. Briefly, differential gene expression was calculated as the difference between the means of control samples and the means of tumor samples (subgroup specific for human samples). In our analysis, we aimed to use controls among platforms that were as similar as possible and selected cerebellar tissue and granule cell precursors as control samples, because these samples were available on every platform compared. This difference was calculated in human samples (d_1) and the murine samples (d_2) separately, and results were represented as a vector in multi-dimensional space. The ADGEX statistic A is the cosine of the angle between d_1 and d_2 in multivariate space and measures the agreement of d_1 and d_2. The statistical significance of the observed AGDEX value (A_{obs}) was determined via testing the null hypothesis that the group assignments in each species (e.g. tumor and control) are exchangeable. By generating 10,000 random permutations, a series of AGDEX values (A_{t1}, A_{t2}, … $A_{t10,000}$) was generated and the p value was defined as the proportion of permutations with a greater absolute agreement value than A_{obs}. p values <0.05 were considered significant. We implemented the analysis in C++ to reduce computing time, compared to other established software packages such as R.

DNA library preparation and sequencing

All gDNA samples, tumors, and respective controls, were processed individually. Where applicable, the purity of the tumor material was tested by thorough histological examination and it was proven that each tumor consisted

of >80 % tumor cells. The Agilent SureSelectXT Mouse All Exon target enrichment kit was used to capture the entire mouse exome for subsequent deep sequencing. The kit targets 50 Mb and covers 221,784 exons within 24,306 gene target regions, which is approximately 1.6 % of the mouse genome. Capturing was performed using a modified SureSelectXT protocol v1.4.1 with 4 pooled pre-capture barcoded libraries of 250 ng each. Libraries were generated according to the Illumina TruSeq DNA Sample Preparation Guide rev. C. Briefly, 1 µg of gDNA was sheared with a Covaris S2 to a mean size of 180–200 bp. 1 µg of the 4 pooled libraries was hybridized for 24 h at 65 °C with the SureSelectXT baits. The captured fragments were sequenced as a 4-plex library in 100 bp paired end mode on one lane of an Illumina HiSeq2000.

Preprocessing of sequencing data

The processing of raw sequencing data was basically done as previously described [14], with the procedure being adapted from human to murine backgrounds. The sequencing reads were aligned to the reference genome mm10 (Genome Reference Consortium Mouse Build 38, GRCm38) with the Burrows-Wheeler Alignment Tool BWA (version 0.6.1-tpx, maximum insert size 1 kb, quality threshold read trimming 20) [28]. Before further processing, aligned reads were sorted by chromosomal coordinates using SAMtools (version 0.1.17) [28]. Duplicated reads were identified and excluded from further analysis using Picard tools (version 1.61). For each sample read coverage was estimated after removal of duplicated reads. Reads were also required to be uniquely mapped (mapping quality mapq $\geq$1) and to have a median base quality of 25. Exome sequencing data were subjected to an additional step extracting the reads mapping to the target region and total read coverage was calculated on this region only.

Detection of single nucleotide variants (SNVs)

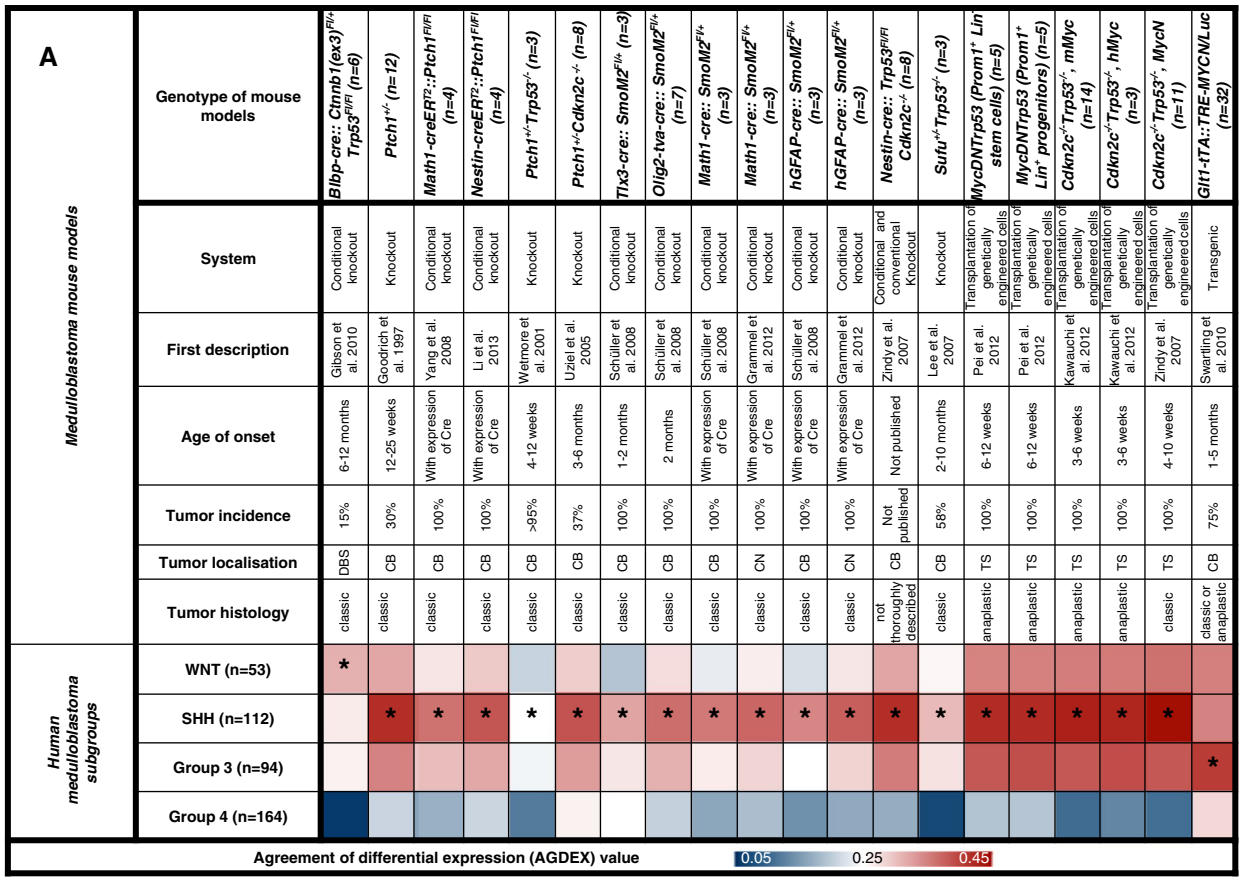
Single nucleotide variants calling was performed analogously to the pipeline described by Jones et al. [14] adapted and summarized as follows: SNVs were initially identified using SAMtools mpileup (version 0.1.17, using the extended base alignment quality BAQ model reducing false SNV calling due to misalignment and high mapping quality reads) [27], and variants were then called on this pileup with SAMtools bcftools (version 0.1.17, probability of variant 2.0), including only reads with high mapping quality and base quality $\geq$13. Additionally, specially tailored filters taking into account coverage, frequency, and surrounding sequence of variants were considered to reduce false positive calls. Variants classified as germline SNVs were excluded, as well as variants that passed the filters only with low confidence. Further analysis was restricted to non-synonymous or nonsense mutations in coding regions only and performed with ANNOVAR [49].

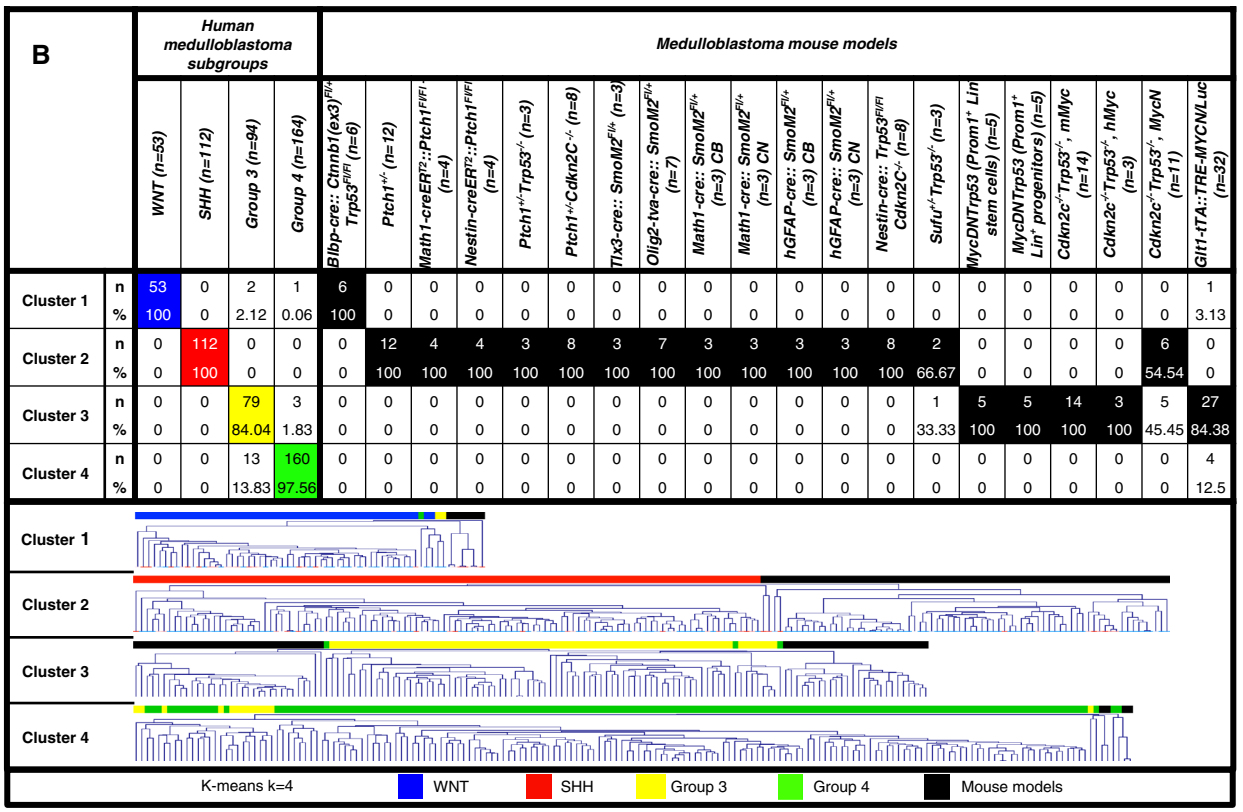
The controls provided a list of candidate mutations. The candidates were matched with genes frequently mutated in human SHH medulloblastoma (19). This procedure was done separately comparing the tumor data to normal data of three different controls and a merged alignment of the latter, respectively. Candidate mutations were also annotated with dbSNP (dbSNP release 137, mouse assembly GRCm38/mm10) to identify potential SNVs as common polymorphisms.

Fig. 1 Characterisation of different medulloblastoma mouse models. **a** Agreement of differential expression (AGDEX) for 20 different medulloblastoma mouse models (140 tumor samples) when compared to the four human medulloblastoma subgroups (423 tumor samples). *Asterisks mark* the highest significant AGDEX score per column ($p < 0.05$). Raw data and all p values can be found in Table S1. **b** *K*-means clustering ($k = 4$) as an independent control method to confirm AGDEX results. Of note, mouse models did not form a distinct cluster but were attributed to the four known subgroups of medulloblastoma which were represented as cluster 1 (WNT subgroup), cluster 2 (SHH subgroup), cluster 3 (Group 3), and cluster 4 (Group 4). A high overlap between AGDEX analyses and *k*-means clustering was seen for the *Blbp-cre::Ctnnb1*$(ex3)^{Fl/+}$*Trp53*$^{Fl/Fl}$ mouse model, for the mouse models based on *Ptch1*, *Smo* or *Sufu* mutations and for the *Glt1-tTA::TRE-MYCN/Luc* mouse model. Mouse models based on *Myc* were attributed to cluster 3. *DBS* dorsal brain stem, *CB* cerebellum, *CN* cochlear nucleus, *TS* transplantation site

Results

Molecular medulloblastoma subtyping may be precisely achieved by global gene expression profiling. In order to thoroughly compare medulloblastoma mouse models to human tumor samples, we analyzed global gene expression data. This was done to (1) further characterize existing mouse models, (2) match these models as precisely as possible to subgroups of human medulloblastoma, and (3) identify mouse models that are most suitable for preclinical studies in a subgroup-specific manner. We collected 140 gene expression datasets from 20 different known medulloblastoma mouse models on the one hand, and gene expression profiles of 423 human medulloblastoma specimens on the other (Fig. 1a). The series of human tumor samples included 53 WNT tumors, 112 SHH tumors, 94 Group 3 tumors, and 164 Group 4 tumors. Epidemiological and clinical data referring to these samples are given in Figure S1. Based on 6996 genes that were present on the murine and human platforms and that were found to be differentially expressed across the four human medulloblastoma subgroups (Figure S2 shows the 25 most significant genes for each subgroup among these differential expressed


A
Medulloblastoma mouse models
Genotype of mouse models
Blbp-cre:: Ctnnb1(ex3)Fl/+ Trp53Fl/Fl (n=6)
Ptch1+/- (n=12)
Math1-creER^T2::Ptch1Fl/Fl (n=4)
Nestin-creER^T2::Ptch1Fl/Fl (n=4)
Ptch1+/-Trp53-/- (n=3)
Ptch1+/-Cdkn2c-/- (n=8)
Tlx3-cre:: SmoM2Fl/+ (n=3)
Olig2-tva-cre:: SmoM2Fl/+ (n=7)
Math1-cre:: SmoM2Fl/+ (n=3)
Math1-cre:: SmoM2Fl/+ (n=3)
hGFAP-cre:: SmoM2Fl/+ (n=3)
hGFAP-cre:: SmoM2Fl/+ (n=3)
Nestin-cre:: Trp53Fl/Fl Cdkn2c-/- (n=8)
Sufu+/-Trp53-/- (n=3)
MycDNTrp53 (Prom1+ Lin- stem cells) (n=5)
MycDNTrp53 (Prom1+ Lin+ progenitors) (n=5)
Cdkn2c-/-Trp53-/-, mMyc (n=14)
Cdkn2c-/-Trp53-/-, hMyc (n=3)
Cdkn2c-/-Trp53-/-, MycN (n=11)
Glt1-tTA::TRE-MYCN/Luc (n=32)
System
Conditional knockout
Knockout
Conditional knockout
Conditional knockout
Knockout
Knockout
Conditional knockout
Conditional knockout
Conditional knockout
Conditional knockout
Conditional knockout
Conditional knockout
Conditional and conventional Knockout
Knockout
Transplantation of genetically engineered cells
Transplantation of genetically engineered cells
Transplantation of genetically engineered cells
Transplantation of genetically engineered cells
Transplantation of genetically engineered cells
Transgenic
First description
Gibson et al. 2010
Goodrich et al. 1997
Yang et al. 2008
Li et al. 2013
Wetmore et al. 2001
Uziel et al. 2005
Schüller et al. 2008
Schüller et al. 2008
Schüller et al. 2008
Schüller et al. 2008
Grammel et al. 2012
Schüller et al. 2008
Grammel et al. 2012
Grammel et al. 2012
Zindy et al. 2007
Lee et al. 2007
Pei et al. 2012
Pei et al. 2012
Kawauchi et al. 2012
Kawauchi et al. 2012
Zindy et al. 2007
Swartling et al. 2010
Age of onset
6-12 months
12-25 weeks
With expression of Cre
With expression of Cre
4-12 weeks
3-6 months
1-2 months
2 months
With expression of Cre
With expression of Cre
With expression of Cre
With expression of Cre
Not published
2-10 months
6-12 weeks
6-12 weeks
3-6 weeks
3-6 weeks
4-10 weeks
1-5 months
Tumor incidence
15%
30%
100%
100%
>95%
37%
100%
100%
100%
100%
100%
100%
Not published
58%
100%
100%
100%
100%
100%
75%
Tumor localisation
DBS
CB
CB
CB
CB
CB
CB
CB
CB
CN
CB
CN
CB
CB
TS
TS
TS
TS
TS
CB
Tumor histology
classic
classic
classic
classic
classic
classic
classic
classic
classic
classic
classic
classic
not thoroughly described
classic
anaplastic
anaplastic
anaplastic
anaplastic
classic
classic or anaplastic
Human medulloblastoma subgroups
WNT (n=53)
SHH (n=112)
Group 3 (n=94)
Group 4 (n=164)
Agreement of differential expression (AGDEX) value
0.05
0.25
0.45


B
Human medulloblastoma subgroups
Medulloblastoma mouse models
WNT (n=53)
SHH (n=112)
Group 3 (n=94)
Group 4 (n=164)
Blbp-cre:: Ctnnb1(ex3)Fl/+ Trp53Fl/Fl (n=6)
Ptch1+/- (n=12)
Math1-creER^T2::Ptch1Fl/Fl (n=4)
Nestin-creER^T2::Ptch1Fl/Fl (n=4)
Ptch1+/-Trp53-/- (n=3)
Ptch1+/-Cdkn2C-/- (n=8)
Tlx3-cre:: SmoM2Fl/+ (n=3)
Olig2-tva-cre:: SmoM2Fl/+ (n=7)
Math1-cre:: SmoM2Fl/+ (n=3) CB
Math1-cre:: SmoM2Fl/+ (n=3) CN
hGFAP-cre:: SmoM2Fl/+ (n=3) CB
hGFAP-cre:: SmoM2Fl/+ (n=3) CN
Nestin-cre:: Trp53Fl/Fl Cdkn2C-/- (n=8)
Sufu+/-Trp53-/- (n=3)
MycDNTrp53 (Prom1+ Lin- stem cells) (n=5)
MycDNTrp53 (Prom1+ Lin+ progenitors) (n=5)
Cdkn2c-/-Trp53-/-, mMyc (n=14)
Cdkn2c-/-Trp53-/-, hMyc (n=3)
Cdkn2c-/-Trp53-/-, MycN (n=11)
Glt1-tTA::TRE-MYCN/Luc (n=32)
Cluster 1 n 53 0 2 1 6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
Cluster 1 % 100 0 2.12 0.06 100 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 3.13
Cluster 2 n 0 112 0 0 0 12 4 4 3 8 3 7 3 3 3 3 8 2 0 0 0 0 6 0
Cluster 2 % 0 100 0 0 0 100 100 100 100 100 100 100 100 100 100 100 100 66.67 0 0 0 0 54.54 0
Cluster 3 n 0 0 79 3 0 0 0 0 0 0 0 0 0 0 0 0 0 1 5 5 14 3 5 27
Cluster 3 % 0 0 84.04 1.83 0 0 0 0 0 0 0 0 0 0 0 0 0 33.33 100 100 100 100 45.45 84.38
Cluster 4 n 0 0 13 160 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 4
Cluster 4 % 0 0 13.83 97.56 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 12.5
Cluster 1
Cluster 2
Cluster 3
Cluster 4
K-means k=4
WNT
SHH
Group 3
Group 4
Mouse models

genes), we performed agreement of differential expression (AGDEX) statistics [12, 38]. AGDEX integrates differential expression analysis results from different platforms and/or species with a higher AGDEX value (maximum value +1) indicating a better agreement between two different groups. We validated our AGDEX results using *k*-means clustering as an independent method that does not rely on control tissue. Details on the mouse models analyzed are provided in Fig. 1a.

Blbp-cre::Ctnnb1$(ex3)^{Fl/+}$*Trp53*$^{Fl/Fl}$ tumors [8] matched best and significantly to the human group of 53 WNT medulloblastomas using both AGDEX analysis (Fig. 1a) and *k*-means clustering ($k = 4$) (Fig. 1b). A small subset of human WNT medulloblastomas (4 out of 20 sequenced cases) did indeed carry a *TP53* mutation in our series. These four tumors did not cluster together or showed significant differences in supervised analysis based on global gene expression (Figure S3).

Using AGDEX analysis, tumors of all analyzed *Patched1* model systems [9, 29, 48, 51, 52, 54] had the highest and significant agreement to the human SHH subgroup (Fig. 1a). This was also true, when looking at these tumors by *k*-means clustering (Fig. 1b). Both statistical methods further showed that all analyzed medulloblastomas driven by an activating mutation in *Smoothened* (*Smo*) [10, 41] matched best and significantly to the human SHH medulloblastoma group (Fig. 1a, b). Eight analyzed *Nestin-cre::Trp53*$^{Fl/Fl}$*Cdkn2c*$^{-/-}$ brain tumors [15, 54] did also match to the SHH subgroup of human medulloblastoma in both analyses, while results for medulloblastoma-like lesions based on a loss of Suppressor of fused (*Sufu*) in a *Trp53* null background [24] were less clear. AGDEX analysis revealed a significant match to the SHH tumors (Fig. 1a), but *k*-means clustering identified one out of three tumors that clustered together with human Group 3 medulloblastoma (Fig. 1b).

Embryonal tumors formed upon transplantation of *Myc*^{T58A} (stabilized Myc) and *DNTrp53* (dominant negative Trp53) transfected cerebellar stem or progenitor cells [37] agreed highly, but not specifically to human Group 3 medulloblastomas using AGDEX analyses (Fig. 1a). Surprisingly, the greatest agreement was found for human SHH medulloblastomas. However, independently performed *k*-means clustering attributed this mouse models to the Group 3 cluster of human medulloblastoma (Fig. 1b). Similar disagreement between the AGDEX results and the *k*-means clustering was seen for other tumor models based on mouse *Myc* (*mMyc*) or human *Myc* (*hMyc*) transfected *Cdkn2c*$^{-/-}$*Trp53*$^{-/-}$ granule neuron precursors, which generate tumors when re-transplanted into murine hosts [15] (Fig. 1a, b).

Transplanted tumors of *MycN* transfected *Cdkn2c*$^{-/-}$ *Trp53*$^{-/-}$ granule neuron precursors [15, 54] agreed best and significantly with human SHH tumors in the AGDEX analysis, and the majority was also attributed to this group via *k*-means clustering (Fig. 1a, b). Finally, transgenic *Glt1-tTA::TRE-MYCN/Luc* mice [45] agreed best and significantly with Group 3 tumors in the AGDEX, as well as the *k*-means-clustering analyses (Fig. 1a, b). Raw data and all *p* values related to Fig. 1a are provided in Table S1.

In summary, based on the analyses performed in the frame of this study, the *Blbp-cre::Ctnnb1*$(ex3)^{Fl/+}$*Trp53*$^{Fl/Fl}$ mouse model appears suitable for the study of WNT medulloblastoma. Mouse models based on *Ptch1* and *Smo* mutations significantly match to human SHH medulloblastoma. *Myc*- or *Mycn*-based mouse models with additional *Trp53* alterations show a SHH signature to some extent, and only the *Glt1-tTA::TRE-MYCN/Luc* mouse model appeared to match clearly to Group 3 tumors. Finally, no mouse models are currently available for Group 4 medulloblastoma.

Most SHH medulloblastomas occur in infants or in adults [17] with distinct gene expression patterns, clinical parameters, and genetic features for each age group [31, 34, 50]. We questioned whether the variety of mouse models that we found to match to human SHH medulloblastoma represented specific models for infant or for adult human SHH cases. First, we confirmed that infants ($\leq$3 years, $n = 29$) and adults ($\geq$16 years, $n = 54$) showed distinct gene expression profiles in our cohort of SHH patients. We identified 705 significantly differentially expressed genes by supervised analysis that were used for the subsequent analyses (out of 10,333 genes that are expressed in both human and murine datasets). Childhood SHH medulloblastomas (between 4 and 15 years of age) did not cluster to a separate subgroup in our cohort, but were more similar to adult SHH medulloblastoma (Figure S4), confirming results from previous studies [31, 50]. Comparison of the significantly differentially expressed genes with genes found in the initial study by Northcott et al. [31] revealed a significant overlap between the two analyses (Figure S5). Interestingly, both studies found several genes of the *HOXA* cluster to be upregulated in adult SHH medulloblastomas (Figure S5).

Most surprisingly, of the 13 different SHH mouse models analyzed, AGDEX analyses revealed 11 models with a high and significant match to human adult SHH patients, while none of the mouse models agreed well with infant SHH medulloblastoma patients (Fig. 2a, exact AGDEX values and *p* values are reported in Table S2). Moreover, *k*-means-clustering ($k = 2$) showed an "infant" and an "adult" SHH cluster, with all analyzed SHH mouse models clustering to the "adult" group (Fig. 2b). Indeed, 287 genes were similarly expressed in murine medulloblastoma models and human adult SHH medulloblastomas and were significantly different from human infant SHH medulloblastomas (Figure S6 and Table S3). Among these genes, several

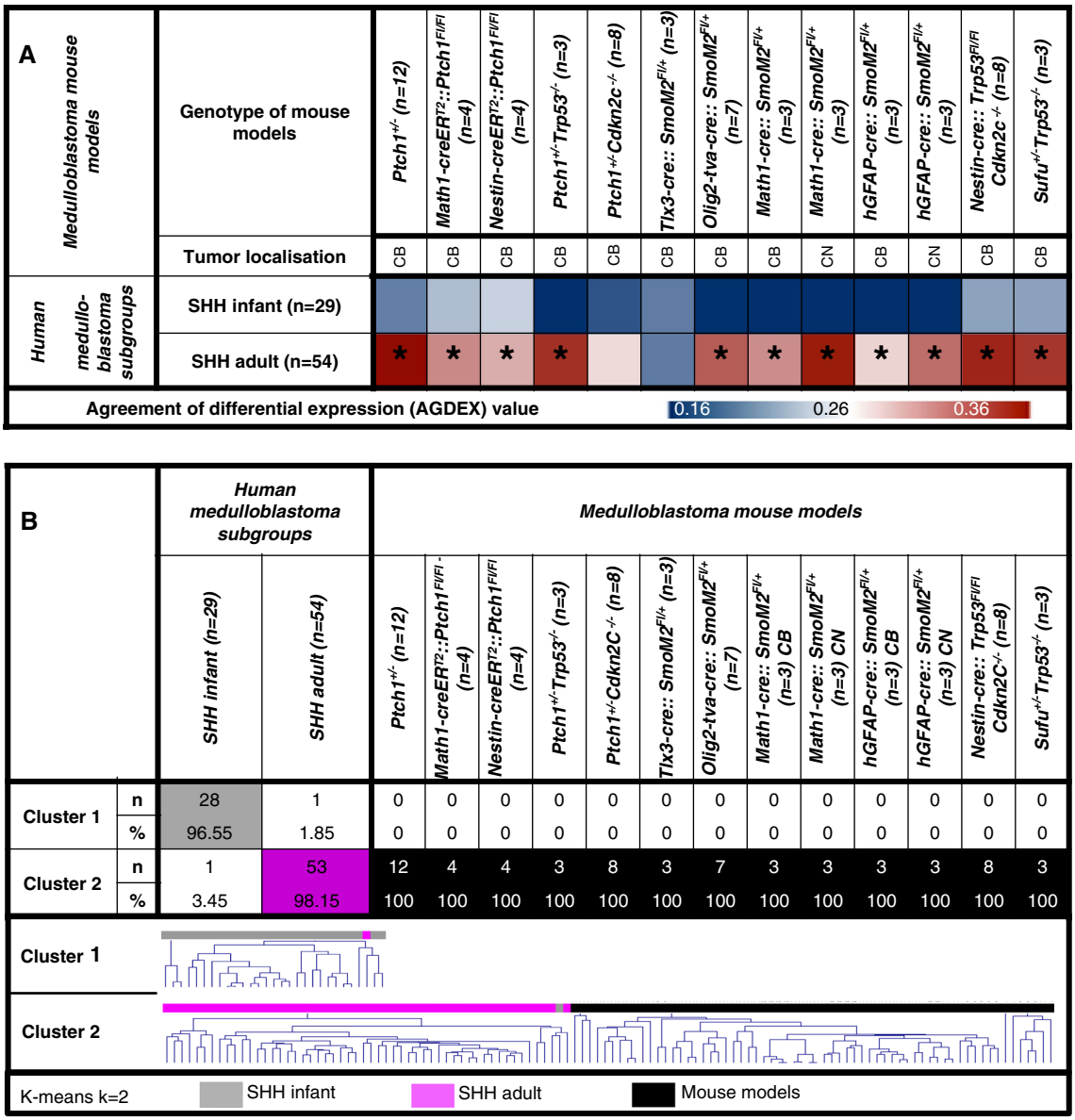
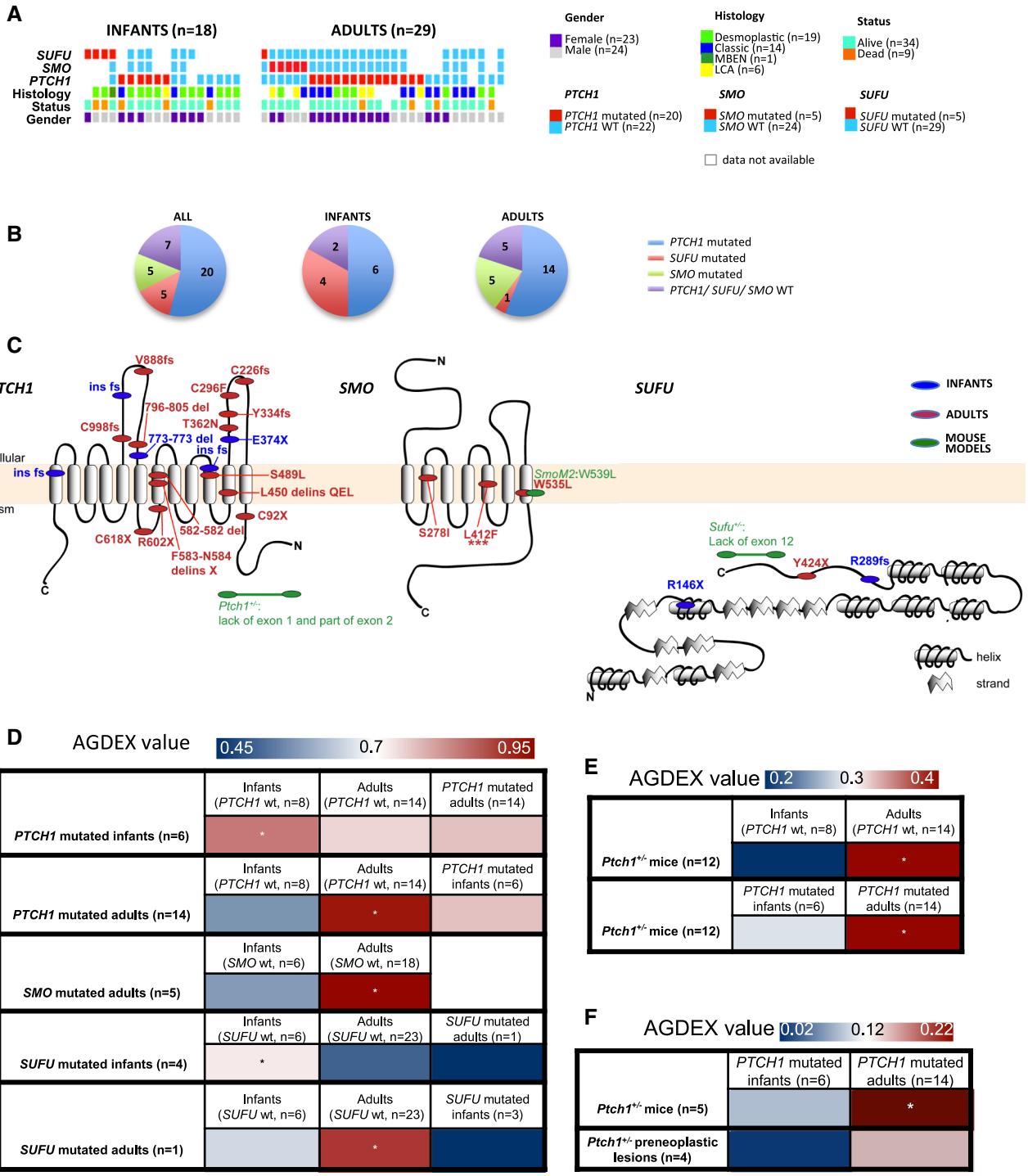


Fig. 2 Mouse models for SHH medulloblastoma display similarities to tumors from adult human patients. **a** Agreement of differential expression (AGDEX) show that mouse models based on *Ptch1*, *SmoM2, Sufu* or *Cdkn2c/Trp53* mutations had a higher agreement with adult human SHH medulloblastoma cases (≥16 years) than to infantile human SHH medulloblastoma cases (≤3 years). *Asterisks* mark the highest significant AGDEX score per column ($p < 0.05$). Raw AGDEX values and all p values can be found in table S2. **b** Confirmation of results by k-means clustering ($k = 2$). All mouse models were sorted to the "adult" cluster of SHH medulloblastoma (cluster 2). *CB* cerebellum, *CN* cochlear nucleus

HoxA genes were upregulated in murine medulloblastoma, similar to the situation in human adult SHH medulloblastomas (Table S3).

One possibility why the existing mouse models matched the adult group of human SHH medulloblastoma was that the driving mutation per se or the global genetic make-up of a tumor determines its gene expression profile. This would mean that, independent of the targeted cell of tumor origin, a *Ptch1* or a *Smo* mutation would always result in a tumor resembling adult human SHH medulloblastoma. Alternatively, murine tumors would over time acquire additional mutations which are typically found in adult SHH medulloblastomas and which would hence determine the respective gene expression profile. In order to evaluate these options, we carefully analyzed our cohort of human SHH tumors with an infantile and with an adult gene expression profile.

Sequencing data for *SUFU, SMO* or *PTCH1* were available for 47 tumors that had previously been characterized with respect to their gene expression profile.



Clinical, histopathological, and genetic characteristics of human infant and adult SHH medulloblastomas are depicted in Fig. 3a. Importantly, mutations in components of the SHH pathway, which have been used to generate many of the SHH mouse models, were distributed unevenly among infants and adults (Fig. 3a, b). *PTCH1* mutations were present in both age groups (6/14 sequenced infantile cases and 14/28 adult cases, χ^2: $p = 0.732$), while *SUFU* mutations were predominantly found in infants (4/10 infant and 1/24 adult cases, $\chi^{2:}$ $p = 0.007$) and *SMO* mutations were exclusively present in adults (0/6 infant and 5/23 adult cases, $\chi^{2:}$ $p = 0.096$). These findings are well in line with previous studies [3, 19]. We did not detect obvious differences between the position or the kind of mutation in the respective genes in infants versus adults (Fig. 3c).

Fig. 3 Distinct mutations in SHH pathway genes do not solely determine an "infantile" or "adult" gene expression signature. **a** Overview on infant and adult SHH medulloblastoma cases that are clearly distinct according to their gene expression profile. *Color codes* display clinical information on patients and mutation status of the SHH pathway genes *SUFU, SMO* and *PTCH1* in human tumors. *LCA* large cell anaplastic, *MBEN* medulloblastoma with extensive nodularity, *WT* wild type. **b** *PTCH1* mutations were detectable in tumor samples from infant and adult patients and *SUFU* mutations were found predominantly in samples from infantile cases. *SMO* mutations were only detected in adult SHH medulloblastoma. **c** Scheme of the transmembrane proteins PTCH1 and SMO, as well as the cytoplasmatic protein SUFU with positions of and details on found mutations in infant SHH medulloblastomas (*blue*) and adult SHH medulloblastomas (*red*). Positions of the respective proteins that are targeted in the *Ptch1*$^{+/-}$, *SmoM2,* and *Sufu*$^{+/-}$ based mouse models are transferred to human proteins and shown in *green*. **d** Independent of their mutations in *PTCH1, SMO,* and *SUFU*, infant and adult SHH medulloblastomas matched the infant and the adult SHH gene expression signature, respectively. Agreement of differential expression (AGDEX) showed that infantile cases with *PTCH1* mutations had a higher agreement with infantile *PTCH1* wt cases than with adult *PTCH1* wt or *PTCH1*-mutated cases ($p = 0.0067$), whereas adult *PTCH1*-mutated cases had the highest agreement with adult *PTCH1* wt cases ($p = 0.004$). Similarly, infantile cases with *SUFU* mutations had a higher agreement with the infantile *SUFU* wt cases than with adult *SUFU* wt or *SUFU*-mutated cases ($p = 0.032$), whereas the adult *SUFU*-mutated tumors matched best to adult *SUFU* wt cases ($p < 0.0001$). Adult *SMO* mutated tumors matched best to adult *SMO* wt cases ($p < 0.0001$). Note that there were no infantile cases with *SMO* mutations (see **a**). *Asterisks* mark the highest and significant AGDEX score per row ($p < 0.05$). **e** Murine *Ptch1*$^{+/-}$ driven tumors significantly matched human tumors from adult patients, independent of their *PTCH1* mutation status ($p = 0.031$ for the match to *PTCH1* wt adults and $p = 0.013$ for the match to *PTCH1*-mutated adults). *Asterisks* mark the highest and significant AGDEX score per row ($p < 0.05$). **f** An independent gene expression array confirmed that murine *Ptch1*$^{+/-}$ driven tumors significantly matched human tumors from adult patients with *PTCH1* mutations ($p = 0.0341$). Even preneoplastic *Ptch1*$^{+/-}$ lesions matched better to adult SHH cases with *PTCH1* mutations than to infant cases with *PTCH1* mutations. ($p = 0.0696$). *Asterisks* in **d**, **e** and **f** mark the highest and significant AGDEX score per row ($p < 0.05$)

In this context, it might not be surprising that the analyzed *SmoM2* mouse models matched human adult SHH medulloblastomas (Fig. 2). However, it appears less clear why models with *Ptch1* mutations behave similarly, and we questioned whether different mutations in the SHH pathway genes may or may not significantly influence an "infant" or "adult" gene expression profile in human SHH medulloblastoma. Using AGDEX statistics, we found that medulloblastomas with *PTCH1* mutations in infants agreed significantly and much better to infantile *PTCH1* wild-type tumors than to adult tumors with *PTCH1* mutations (Fig. 3d). Conversely, *PTCH1*-mutated medulloblastomas from adults significantly agreed with adult *PTCH1* wild-type tumors, but not with infantile *PTCH1*-mutated medulloblastomas (Fig. 3d). Analogous results were seen for *SUFU*-mutated tumors from infants and adults, respectively (Fig. 3d). While infants with *SMO* mutations were not identified in our series, adult *SMO*-mutated medulloblastomas agreed well with adult wild-type *SMO* tumors (Fig. 3d).

In an additional experiment we looked at murine *Ptch1*$^{+/-}$ driven tumors, which we assumed were genetically more homogeneous than human medulloblastoma with *PTCH1* mutations. As depicted in Fig. 2a, such *Ptch1*$^{+/-}$ driven tumors significantly matched human tumors from adult patients. Intriguingly, this was still true when selectively comparing the murine tumors to human *PTCH1* wild-type cases, and it was also true when comparing them to *PTCH1*-mutated cases only (Fig. 3e). At the same time, the AGDEX values for the adult *PTCH1* wt cases (0.408) and the adult *PTCH1*-mutated cases (0.401) were nearly identical, when comparing these samples to murine *Ptch1*$^{+/-}$ medulloblastomas (Fig. 3e). Together, these results suggest that mutations in SHH pathway genes do not have a significant influence on the "infant" or "adult" gene expression profile of a SHH tumor, and they do not even influence the gene expression profile in a significant way, when only looking at adult SHH medulloblastomas.

Next, we questioned whether additionally acquired mutations may be responsible for the adult fingerprint of SHH medulloblastoma. This appeared somewhat likely, since adult SHH medulloblastoma usually carry significantly more mutations than infant cases [19]. As a first step to address this question, we analyzed the expression profiles of preneoplastic *Ptch1*$^{+/-}$ lesions, which are an early event of medulloblastoma generation in *Ptch1*$^{+/-}$ mice [36]. Assuming that such early lesions would not yet have acquired the full spectrum of additional "bystander" mutations, we asked whether they still matched better to adult SHH cases with *PTCH1* mutations than to infant cases with *PTCH1* mutations. As demonstrated in Fig. 3f, the AGDEX value was high when comparing preneoplastic *Ptch1*$^{+/-}$ lesions and adult human SHH medulloblastoma. However, results were not statistically significant confirming that these preneoplastic *Ptch1*$^{+/-}$ lesions do not represent a fully developed medulloblastoma yet.

We, therefore, asked whether full-blown medulloblastoma from *Ptch1*$^{+/-}$ mice carried additional mutations that are typical of adult human SHH medulloblastoma. To this end, we screened 20 human infant *PTCH1*-mutated SHH tumors and 29 adult *PTCH1*-mutated SHH medulloblastomas for additional mutations and selected 29 genes that were repeatedly ($\geq$2 cases) and specifically mutated in adult *PTCH1*-mutated SHH tumors (Table 1, [19]). In order to answer if these additional mutations contributed to the "adult" gene expression profile in murine *Ptch1*$^{+/-}$ tumors, we performed whole exome sequencing and analyzed the corresponding genes in our murine samples. We confirmed the loss of *Patched1* in our mouse model, but no

Table 1 Murine *Ptch1*$^{+/-}$ tumors do not display additional mutations that are repeatedly found in human adult *PTCH1*-mutated SHH medulloblastomas

Gene symbol	Infants with *PTCH1* mutation ($n = 20$)	Adults with *PTCH1* mutation ($n = 29$)	*Ptch1* ± mice ($n = 4$)
PTCH1	20 (100 %)	29 (100 %)	4 (100 %)
DDX3X	0 (0 %)	12 (41 %)	0 (0 %)
CREBBP	0 (0 %)	4 (14 %)	0 (0 %)
KIAA0182	0 (0 %)	4 (14 %)	0 (0 %)
LDB1	1 (5 %)	4 (14 %)	0 (0 %)
MLL2	0 (0 %)	4 (14 %)	0 (0 %)
TCF4	0 (0 %)	4 (14 %)	0 (0 %)
PCDH19	0 (0 %)	3 (10 %)	0 (0 %)
ABCA5	0 (0 %)	3 (10 %)	0 (0 %)
ARHGAP28	0 (0 %)	2 (7 %)	0 (0 %)
C12ORF40	0 (0 %)	2 (7 %)	0 (0 %)
CCAR1	0 (0 %)	2 (7 %)	0 (0 %)
DCAF12L2	0 (0 %)	2 (7 %)	0 (0 %)
FAM47C	0 (0 %)	2 (7 %)	0 (0 %)
FBXW7	0 (0 %)	2 (7 %)	0 (0 %)
KDM3B	0 (0 %)	2 (7 %)	0 (0 %)
KIAA1267	0 (0 %)	2 (7 %)	0 (0 %)
KIF26B	0 (0 %)	2 (7 %)	0 (0 %)
LRP1B	0 (0 %)	2 (7 %)	0 (0 %)
MLL	0 (0 %)	2 (7 %)	0 (0 %)
MLL3	0 (0 %)	2 (7 %)	0 (0 %)
NEB	0 (0 %)	2 (7 %)	0 (0 %)
NSD1	0 (0 %)	2 (7 %)	0 (0 %)
PTEN	0 (0 %)	2 (7 %)	0 (0 %)
PTPRD	0 (0 %)	2 (7 %)	0 (0 %)
SMARCA2	0 (0 %)	2 (7 %)	0 (0 %)
SYNE1	0 (0 %)	2 (7 %)	0 (0 %)
UPK3B	0 (0 %)	2 (7 %)	0 (0 %)
UVRAG	0 (0 %)	2 (7 %)	0 (0 %)
WDR47	0 (0 %)	2 (7 %)	0 (0 %)

20 Human infant *PTCH1*-mutated SHH tumors and 29 adult *PTCH1*-mutated SHH medulloblastomas were screened for additional mutations and 29 genes that were repeatedly (≥2 cases) and specifically mutated in adult *PTCH1*-mutated SHH tumors were selected. In murine *Ptch1*$^{+/-}$ tumors, loss of *Patched* was confirmed, but no adult-specific mutations were detected

adult-specific mutations were detected (Table 1). Indeed, exome sequencing did not reveal any recurrent mutations in our murine *Ptch1*$^{+/-}$ tumor samples. Similarly, adult human *SMO*-mutated medulloblastomas display multiple additional mutations [19], but we did not detect any additional mutation in four cerebellar and three cochlear tumors of the *hGFAP-cre::SmoM2*$^{FL/+}$ mice. Finally, we also looked at copy number aberrations in 7 *Ptch1*$^{+/-}$ and in 7 *Math1-cre::SmoM2*$^{FL/+}$ medulloblastoma to see whether there are gross chromosomal abnormalities that would be comparable to the changes in human cases. However, mouse tumor genomes appeared rather stable apart from a trisomy of chromosome 6 in some cases (data not shown), which is well in line with previously published findings [24, 54]. In summary, these results clearly argue against a scenario where a tumor's driver mutation or the global genetic make-up solely determines the gene expression profile of a tumor.

Discussion

We integrated global gene expression profiles of the majority of available medulloblastoma mouse models with a large cohort of human medulloblastoma patients. Based on the current consensus of four distinct human molecular subgroups of medulloblastoma [47] we show that the present WNT mouse model (*Blbp-cre::Ctnnb1*(*ex3*)$^{Fl/+}$*Trp53*$^{Fl/Fl}$) is suitable for the study of the respective human medulloblastoma subgroup, independent of the *TP53* mutation status. Our data, therefore, confirm and extend the results of the initial study by Gibson and colleagues, where this model was demonstrated to match to a group of 6 human WNT tumors [8].

The *MycDNTrp53* models were initially presented as Group 3 models using a 50-gene panel to identify non-SHH/WNT tumors and by an extra 100-gene panel to distinguish between Groups 3 and 4. These models, as well

as *Cdkn2c*$^{-/-}$*Trp53*$^{-/-}$*, hMyc or mMyc* models [15, 37] did not match specifically to human Group 3 medulloblastomas, but agreed somewhat with human SHH medulloblastoma in the ADGEX analysis. These discrepancies may on the one hand be explained by the fact that our analysis relied on global gene expression profiles and not on limited gene panels. On the other hand, the similarity to the SHH group may be caused by the *TP53* alterations that are part of these models. Such mutations occur in SHH medulloblastoma, but have not been detected in any of the 72 human Group 3 medulloblastoma that we have recently analyzed [53]. The mismatching may also be related to artificial alterations that may have been acquired in vitro prior to the transplantation of the tumor cells. More importantly, the targeted cells used in the studies may not exactly reflect the cell of origin for human Group 3 tumors. While the potential cell of origin has not been identified yet, we recently suggested that Group 3 tumors may have an extracerebellar origin [50].

We found that the *Cdkn2c*$^{-/-}$*Trp53*$^{-/-}$, *MycN* model matched best to SHH medulloblastoma, which is in line with initial findings [15]. *MYCN* amplifications are not only present in human Group 3 and 4 medulloblastoma [39], but also in SHH medulloblastomas, and *MYCN* is a known target of the SHH signaling pathway, which is expressed in almost all SHH medulloblastoma. Interestingly, the *Glt1-tTA::TRE-MYCN/Luc* mouse model [45] was the only model with a clear match to human Group 3 tumors in both statistical analyses, possibly due to the fact that this model is a genetically engineered organism that does not include any in vitro steps and that it is not relying on a *Trp53* alteration. Importantly, our results reveal a total lack of Group 4 mouse models, since none of our analyzed murine tumors had similarities with human Group 4 medulloblastomas.

We certainly acknowledge that our comparison of mouse and human medulloblastoma is solely based on global gene expression patterns. AGDEX and k-means clustering performed here represent independent statistical methods—while AGDEX analyses are based on means and differential gene expression in comparison to controls, *k*-means and Pearson correlation simply correlate gene expression values of each sample and no controls are needed. Due to these differences, results of these analyses were not always concordant in our study, underlining that independent statistical methods are crucial to exactly align a mouse model to the human tumor. Strictly, when it comes to evaluate a tumor model, we believe that concordance of different statistical analyses of gene expression together with other factors, such as histology, aggressiveness, and metastatic behavior, has to be considered. Moreover, the utility of a model definitely depends on the purpose for which it is used and can only be determined once the findings from a given model are applied to the appropriate clinical purpose. Nevertheless, global gene expression profiling initially established the now acknowledged four subgroups of medulloblastoma and, therefore, needs to be regarded as a major criterion in mouse models for medulloblastoma.

The SHH subgroup of human medulloblastoma shows a two-peak age distribution with many tumors occurring in infants or in adult patients, and these tumors display distinct gene expression patterns as well as different clinical parameters, genetic features, and locations [31, 50]. Our results suggest that Shh-associated mouse models, based on *Ptch1* or *Smo* mutations, are adequate to study adult human SHH medulloblastoma cases, since their gene expression profile significantly matched the human counterparts. Surprisingly, *Sufu*$^{+/-}$*Trp53*$^{-/-}$ tumors do not exactly model infant SHH medulloblastomas, although *SUFU* mutations almost exclusively occur in infants with SHH medulloblastoma. Moreover, *Sufu*$^{+/-}$*Trp53*$^{-/-}$ tumors do not display desmoplastic features, whereas *SUFU* mutations are clearly associated with desmoplastic/nodular histology in humans (see Fig. 3a, [3]). One possible reason may be represented by the *Trp53* mutation of the model that is usually not detected in human infant tumors [19]. Since *Sufu*$^{-/-}$ mice are not viable and *Sufu*$^{+/-}$ mice with a *Trp53* wild-type situation do not develop medulloblastoma [24], a faithful model for infant SHH medulloblastoma is still to be generated. While a homozygous deletion of *Sufu* may technically be achievable by a conditional knockout, such models both need to harbor the right composition of oncogenic mutations and need to target the right cell type. In fact, mice with a homozygous loss of *Sufu* under the control of the *Hoxb.7* promoter, which is active 'in all cells of the developing cerebellum,' develop severe cerebellar mispatterning, but no medulloblastoma [16]. Therefore, mouse models for infant SHH medulloblastoma are urgently needed, not only to better understand tumor biology, but also to be able to conduct preclinical studies with an authentic model. Currently, treatment studies on murine SHH medulloblastoma models have to be regarded with caution, and successful drugs might not be easily applicable to infant patients.

The mutational spectrum is significantly different in infant and adult SHH medulloblastoma. Moreover, we demonstrate that infant and adult human SHH medulloblastomas display an "infant" or "adult" gene expression profile, respectively, but this was independent of their mutation in SHH pathway genes. *Ptch1*$^{+/-}$ mice clearly matched better to human adult SHH tumors without *PTCH1* mutations than to human infant tumors with *PTCH1* mutations. Apart from different frequencies in SHH pathway gene mutations, adult SHH medulloblastomas display a lot of additional "bystander" mutations that are not seen in infant SHH medulloblastomas [19]. In our study, *Ptch1*$^{+/-}$ mice showed an adult SHH gene expression pattern, but did not display additional mutations that are typically found in

human adult SHH tumors. Hence, the mutation pattern *per se* is not the main determinant of the "adult" gene expression profile in SHH medulloblastoma.

Our data strongly suggest that multiple features beyond tumor genetics have to be considered as driving forces for a gene expression profile, and epigenetic alterations, for instance, appear pretty likely in this context. Interestingly, we and others found several genes of the *HOXA* cluster to be upregulated in human adult SHH medulloblastoma [31]. Similarly, *Ptch1* and *Smo* mouse models showed an upregulation of *HoxA* genes. The *HOXA* cluster belongs to the *HOX* gene family that encodes homeodomain-containing transcription factors, which play important roles in mammalian development (for review see [7]). Epigenetic control of *HOXA* gene expression via methylation was described in several in vitro studies [1, 2, 23], and *Hoxa5* tissue-specific methylation patterns were detected during murine development [11]. Moreover, hypermethylation of *HOXA* genes, with respective aberrant expression, was found in various tumors, such as breast cancer, leukemia or lung carcinoma [20, 35, 42, 44]. The *HOXA* genes are also known methylation targets in other brain tumors such as gliomas or meningiomas, where they were described as prognostic markers [4, 5]. Hence, differential expression of several *HOXA* genes might be one of the features attributed to the striking differential methylation patterns that have recently been discovered in infant and adult SHH medulloblastomas [19].

Finally, infant and adult SHH medulloblastomas might arise from a different cell of origin that substantially influences the respective gene expression profile. For instance, mice with oncogenic Shh mutations driven by *Math1* or *hGFAP* promoter sequences generate medulloblastomas from granule neuron precursors, and these models matched excellent to human adult SHH tumors. Similarly, Nestin-positive cells within the external granule cell layer generate tumors that match significantly to human adult SHH medulloblastoma. Hence, cells within the external granule cell layer may be recognized as the cell of origin for adult human SHH medulloblastoma, whereas the cell of origin for infant SHH medulloblastoma remains to be elucidated and may eventually lie outside the EGL. These ideas are supported by the fact that human infant and adult SHH medulloblastomas show clearly distinct locations within the hindbrain. While infant SHH medulloblastomas show association to the brainstem, the majority of adult SHH medulloblastomas grow exclusively in the cerebellum [50].

In summary, we show here that the genetic make-up of a tumor is not the main determinant of its gene expression profile and that other factors such as epigenetics and the cell of origin must play important roles. Consequently, molecular subgrouping based on gene expression, which is nowadays being performed for a variety of tumors, may give us—independent of the mutational spectrum—direct hints for different cells of origin or epigenetic features, even within tumors that previously appeared to represent a single entity.

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References

1. Atkinson SP, Koch CM, Clelland GK et al (2008) Epigenetic marking prepares the human HOXA cluster for activation during differentiation of pluripotent cells. Stem Cells 26:1174–1185. doi:10.1634/stemcells.2007-0497
2. Bocker MT, Tuorto F, Raddatz G et al (2012) Hydroxylation of 5-methylcytosine by TET2 maintains the active state of the mammalian HOXA cluster. Nat Commun 3:818. doi:10.1038/ncomms1826
3. Brugieres L, Remenieras A, Pierron G et al (2012) High frequency of germline SUFU mutations in children with desmoplastic/nodular medulloblastoma younger than 3 years of age. J Clin Oncol 30:2087–2093. doi:10.1200/JCO.2011.38.7258
4. Di Vinci A, Brigati C, Casciano I et al (2012) HOXA7, 9, and 10 are methylation targets associated with aggressive behavior in meningiomas. Transl Res 160:355–362. doi:10.1016/j.trsl.2012.05.007
5. Di Vinci A, Casciano I, Marasco E et al (2012) Quantitative methylation analysis of HOXA3, 7, 9, and 10 genes in glioma: association with tumor WHO grade and clinical outcome. J Cancer Res Clin Oncol 138:35–47. doi:10.1007/s00432-011-1070-5
6. Fattet S, Haberler C, Legoix P et al (2009) Beta-catenin status in paediatric medulloblastomas: correlation of immunohistochemical expression with mutational status, genetic profiles, and clinical characteristics. J Pathol 218:86–94. doi:10.1002/path.2514
7. Favier B, Dolle P (1997) Developmental functions of mammalian Hox genes. Mol Hum Reprod 3:115–131
8. Gibson P, Tong Y, Robinson G et al (2010) Subtypes of medulloblastoma have distinct developmental origins. Nature 468:1095–1099. doi:10.1038/nature09587
9. Goodrich LV, Milenkovic L, Higgins KM, Scott MP (1997) Altered neural cell fates and medulloblastoma in mouse patched mutants. Science 277:1109–1113
10. Grammel D, Warmuth-Metz M, von Bueren AO et al (2012) Sonic hedgehog-associated medulloblastoma arising from the cochlear nuclei of the brainstem. Acta Neuropathol 123:601–614. doi:10.1007/s00401-012-0961-0
11. Hershko AY, Kafri T, Fainsod A, Razin A (2003) Methylation of HoxA5 and HoxB5 and its relevance to expression during mouse development. Gene 302:65–72
12. Johnson RA, Wright KD, Poppleton H et al (2010) Cross-species genomics matches driver mutations and cell compartments to model ependymoma. Nature 466:632–636. doi:10.1038/nature09173
13. Johnson WE, Li C, Rabinovic A (2007) Adjusting batch effects in microarray expression data using empirical Bayes methods. Biostatistics 8:118–127. doi:10.1093/biostatistics/kxj037
14. Jones DT, Jäger N, Kool M et al (2012) Dissecting the genomic complexity underlying medulloblastoma. Nature 488:100–105. doi:10.1038/nature11284

15. Kawauchi D, Robinson G, Uziel T et al (2012) A mouse model of the most aggressive subgroup of human medulloblastoma. Cancer Cell 21:168–180. doi:10.1016/j.ccr.2011.12.023
16. Kim JJ, Gill PS, Rotin L et al (2011) Suppressor of fused controls mid-hindbrain patterning and cerebellar morphogenesis via GLI3 repressor. J Neurosci 31:1825–1836. doi:10.1523/JNEUROSCI.2166-10.201131/5/1825
17. Kool M, Korshunov A, Remke M et al (2012) Molecular subgroups of medulloblastoma: an international meta-analysis of transcriptome, genetic aberrations, and clinical data of WNT, SHH, Group 3, and Group 4 medulloblastomas. Acta Neuropathol 123:473–484. doi:10.1007/s00401-012-0958-8
18. Kool M, Koster J, Bunt J et al (2008) Integrated genomics identifies five medulloblastoma subtypes with distinct genetic profiles, pathway signatures and clinicopathological features. PLoS ONE 3:e3088. doi:10.1371/journal.pone.0003088
19. Kool M, Jäger N, Northcott P et al (2014) Genome sequencing of SHH medulloblastoma predicts genotype-related response to smoothened-inhibition. Cancer Cell 25(3):393–405. doi:10.1016/j.ccr.2014.02.004
20. Krivtsov AV, Feng Z, Lemieux ME et al (2008) H3K79 methylation profiles define murine and human MLL-AF4 leukemias. Cancer Cell 14:355–368. doi:10.1016/j.ccr.2008.10.001
21. Lattin JE, Schroder K, Su AI et al (2008) Expression analysis of G Protein-Coupled Receptors in mouse macrophages. Immunome Res 4:5. doi:10.1186/1745-7580-4-5
22. Lau J, Schmidt C, Markant SL, Taylor MD, Wechsler-Reya RJ, Weiss WA (2012) Matching mice to malignancy: molecular subgroups and models of medulloblastoma. Childs Nerv Syst 28:521–532. doi:10.1007/s00381-012-1704-1
23. Laurent L, Wong E, Li G et al (2010) Dynamic changes in the human methylome during differentiation. Genome Res 20:320–331. doi:10.1101/gr.101907.109
24. Lee Y, Kawagoe R, Sasai K et al (2007) Loss of suppressor-of-fused function promotes tumorigenesis. Oncogene 26:6442–6447. doi:10.1038/sj.onc.1210467
25. Leek JT, Johnson WE, Parker HS, Jaffe AE, Storey JD (2012) The sva package for removing batch effects and other unwanted variation in high-throughput experiments. Bioinformatics 28:882–883. doi:10.1093/bioinformatics/bts034
26. Lepourcelet M, Tou L, Cai L et al (2005) Insights into developmental mechanisms and cancers in the mammalian intestine derived from serial analysis of gene expression and study of the hepatoma-derived growth factor (HDGF). Development 132:415–427
27. Li H (2011) Improving SNP discovery by base alignment quality. Bioinformatics 27:1157–1158. doi:10.1093/bioinformatics/btr076
28. Li H, Handsaker B, Wysoker A et al (2009) The sequence alignment/map format and SAMtools. Bioinformatics 25:2078–2079. doi:10.1093/bioinformatics/btp352
29. Li P, Du F, Yuelling LW et al (2013) A population of Nestin-expressing progenitors in the cerebellum exhibits increased tumorigenicity. Nat Neurosci 16:1737–1744. doi:10.1038/nn.3553
30. Mao J, Ligon KL, Rakhlin EY et al (2006) A novel somatic mouse model to survey tumorigenic potential applied to the Hedgehog pathway. Cancer Res 66:10171–10178
31. Northcott PA, Hielscher T, Dubuc A et al (2011) Pediatric and adult sonic hedgehog medulloblastomas are clinically and molecularly distinct. Acta Neuropathol (Berl) 122:231–240
32. Northcott PA, Jones DT, Kool M et al (2012) Medulloblastomics: the end of the beginning. Nat Rev Cancer 12:818–834. doi:10.1038/nrc3410nrc3410
33. Northcott PA, Korshunov A, Witt H et al (2011) Medulloblastoma comprises four distinct molecular variants. J Clin Oncol 29:1408–1414
34. Northcott PA, Shih DJ, Peacock J et al (2012) Subgroup-specific structural variation across 1,000 medulloblastoma genomes. Nature 488:49–56. doi:10.1038/nature11327
35. Novak P, Jensen T, Oshiro MM et al (2006) Epigenetic inactivation of the HOXA gene cluster in breast cancer. Cancer Res 66:10664–10670. doi:10.1158/0008-5472.CAN-06-2761
36. Oliver TG, Read TA, Kessler JD et al (2005) Loss of patched and disruption of granule cell development in a pre-neoplastic stage of medulloblastoma. Development 132:2425–2439
37. Pei Y, Moore CE, Wang J et al (2012) An animal model of MYC-driven medulloblastoma. Cancer Cell 21:155–167. doi:10.1016/j.ccr.2011.12.021
38. Pounds S, Gao CL, Johnson RA et al (2011) A procedure to statistically evaluate agreement of differential expression for cross-species genomics. Bioinformatics 27:2098–2103. doi:10.1093/bioinformatics/btr362
39. Robinson G, Parker M, Kranenburg TA et al (2012) Novel mutations target distinct subgroups of medulloblastoma. Nature 488:43–48. doi:10.1038/nature11213
40. Rutkowski S, von Hoff K, Emser A et al (2010) Survival and prognostic factors of early childhood medulloblastoma: an international meta-analysis. J Clin Oncol 28:4961–4968
41. Schüller U, Heine VM, Mao J et al (2008) Acquisition of granule neuron precursor identity is a critical determinant of progenitor cell competence to form Shh-induced medulloblastoma. Cancer Cell 14:123–134. doi:10.1016/j.ccr.2008.07.005
42. Shiraishi M, Sekiguchi A, Oates AJ, Terry MJ, Miyamoto Y (2002) HOX gene clusters are hotspots of de novo methylation in CpG islands of human lung adenocarcinomas. Oncogene 21:3659–3662. doi:10.1038/sj.onc.1205453
43. Shirasawa S, Arata A, Onimaru H et al (2000) Rnx deficiency results in congenital central hypoventilation. Nat Genet 24:287–290. doi:10.1038/73516
44. Strathdee G, Holyoake TL, Sim A et al (2007) Inactivation of HOXA genes by hypermethylation in myeloid and lymphoid malignancy is frequent and associated with poor prognosis. Clin Cancer Res 13:5048–5055. doi:10.1158/1078-0432.CCR-07-0919
45. Swartling FJ, Grimmer MR, Hackett CS et al (2010) Pleiotropic role for MYCN in medulloblastoma. Genes Dev 24:1059–1072. doi:10.1101/gad.190751024/10/1059
46. Swartling FJ, Savov V, Persson AI et al (2012) Distinct neural stem cell populations give rise to disparate brain tumors in response to N-MYC. Cancer Cell 21:601–613. doi:10.1016/j.ccr.2012.04.012
47. Taylor MD, Northcott PA, Korshunov A et al (2012) Molecular subgroups of medulloblastoma: the current consensus. Acta Neuropathol 123:465–472. doi:10.1007/s00401-011-0922-z
48. Uziel T, Zindy F, Xie S et al (2005) The tumor suppressors Ink4c and p53 collaborate independently with Patched to suppress medulloblastoma formation. Genes Dev 19:2656–2667. doi:10.1101/gad.1368605
49. Wang K, Li M, Hakonarson H (2010) ANNOVAR: functional annotation of genetic variants from high-throughput sequencing data. Nucleic Acids Res 38:e164. doi:10.1093/nar/gkq603gkq603
50. Wefers AK, Warmuth-Metz M, Pöschl J et al (2014) Subgroup-specific localization of human medulloblastoma based on pre-operative MRI. Acta Neuropathol 127:931–933. doi:10.1007/s00401-014-1271-5
51. Wetmore C, Eberhart DE, Curran T (2001) Loss of p53 but not ARF accelerates medulloblastoma in mice heterozygous for patched. Cancer Res 61:513–516
52. Yang ZJ, Ellis T, Markant SL et al (2008) Medulloblastoma can be initiated by deletion of Patched in lineage-restricted progenitors or stem cells. Cancer Cell 14:135–145. doi:10.1016/j.ccr.2008.07.003

53. Zhukova N, Ramaswamy V, Remke M et al (2013) Subgroup-specific prognostic implications of TP53 mutation in medulloblastoma. J Clin Oncol 31:2927–2935. doi:10.1200/JCO.2012.48.5052

54. Zindy F, Uziel T, Ayrault O et al (2007) Genetic alterations in mouse medulloblastomas and generation of tumors de novo from primary cerebellar granule neuron precursors. Cancer Res 67:2676–2684. doi:10.1158/0008-5472.CAN-06-3418